Hotel Revenue and Demand Forecasting

Short project report | Python, pandas, Prophet | Hotel Booking Demand dataset

119,390	2 years	90 days	Prophet
bookings analysed (32 columns)	daily history, Jul 2015 to Aug 2017	forecast horizon (to 29 Nov 2017)	trend + weekly + yearly seasonality

Key insights

1 Demand is seasonal and weekly
Bookings rise in spring and summer and fall in winter. A strong day-of-week cycle sits on top, and the overall level drifts upward from 2015 to 2017.

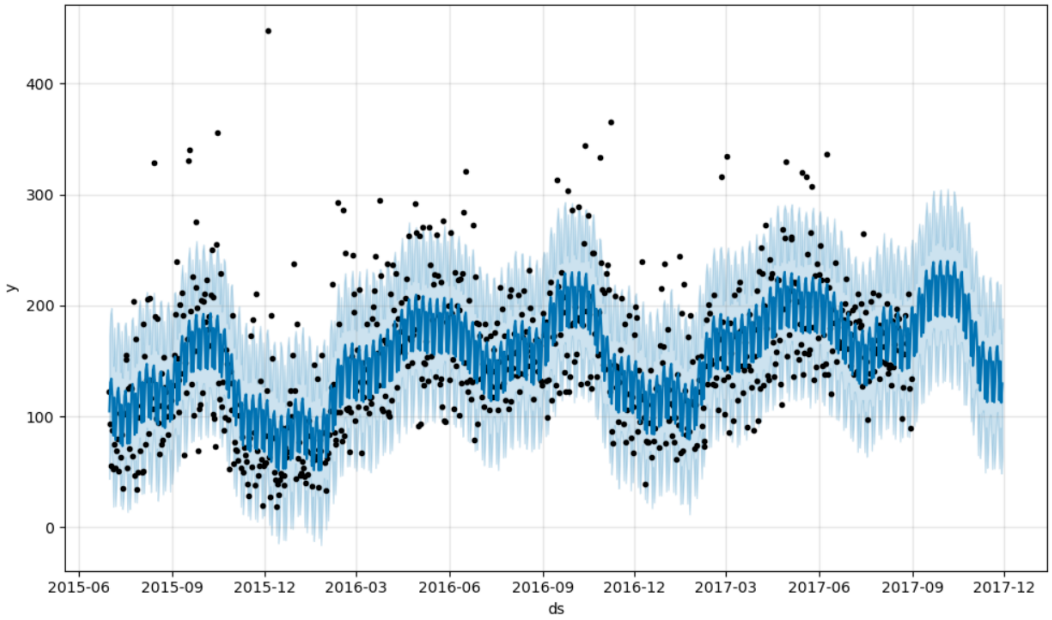
2 Price follows the season
Daily ADR peaks in July and August (about 160 EUR in 2016, about 180 EUR in 2017, read from the chart) and drops to roughly 60 EUR in winter.

3 Revenue tracks ADR and volume
Summer revenue peaks are clearly higher in 2017 than in 2016, so growth came from both rate and volume. Winter days fall to a small fraction of the peak.

4 Cancellations are noisy and growing
Cancellations spike unpredictably (up to about 225 per day) and rise in absolute terms in 2017 as booking volume grows. They deserve their own model.

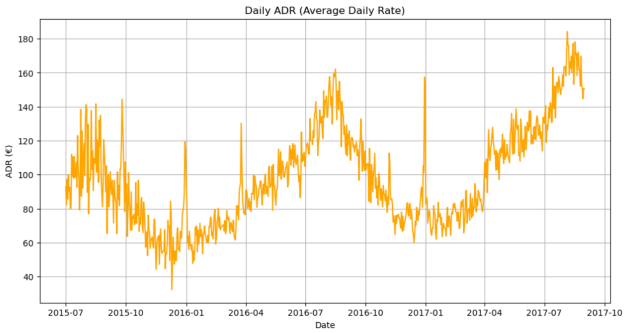
Forecast

The Prophet model fits the history well and projects a seasonal peak around October 2017 followed by the usual winter decline. Prediction intervals are wide: for 29 Nov 2017 the point forecast is about 130 bookings, with an interval of roughly 64 to 188.

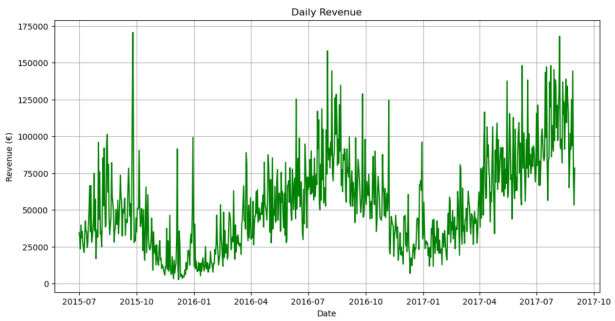


Daily demand: observed bookings (dots), Prophet forecast (line) and uncertainty band.

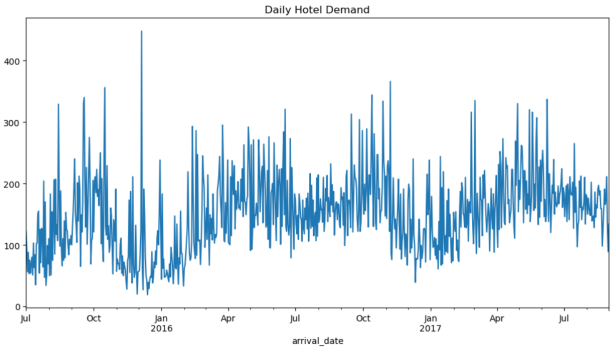
Supporting views



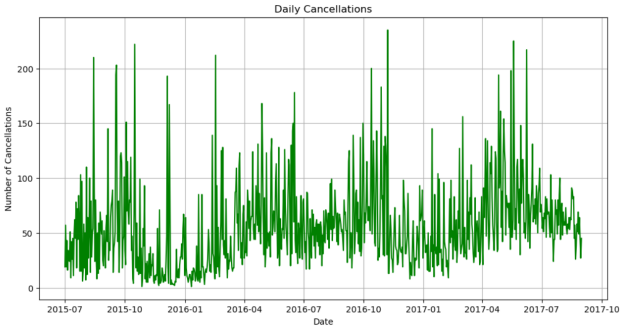
Daily ADR (EUR)



Daily revenue (EUR)



Daily demand (bookings)



Daily cancellations

Limitations and next steps

• **No accuracy check yet.** The model was fitted and plotted but not tested on held-out data, so no MAE or MAPE is reported. Add a time-based holdout (for example, the last 90 days) before relying on the numbers.

• **Demand includes cancelled bookings.** Daily demand counts every booking row, cancelled or not, so it measures booking requests rather than occupied rooms. Revenue is built the same way, so it may be overstated.

• **Both hotels are pooled.** Results are aggregated across the whole dataset and are not split by hotel type.

• **Only about two seasonal cycles.** Yearly seasonality is learned from two summers, which limits how far the forecast can be trusted.

Method: loaded and parsed arrival dates, aggregated bookings, mean ADR, cancellations and revenue by arrival date, fitted Prophet (weekly and yearly seasonality) on daily demand, forecast 90 days.

Source: Hotel Booking Demand dataset (hotel_bookings demand.csv), 119,390 rows.